

University of Wah

Department of Computer Science



Submit to:

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Submit by:

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Course:

Information and Security Lab - CS-251L

Reg No:

UW-24-CS-BS-101

Section:

BS-CS-3rd-B

Original Code

Playfair Matrix:

MONAR

CHYBD

EFGIK

LPQST

UWXZ

Plaintext: INSTRUMENT

Ciphertext: GATLMZCLRQ

Decrypted: INSTRUMENT

Task Pages

DAY: _____

DATE: _____

Task # 06

Detail:

Name : TALHA

Fav Game : CRICKET

Key : SECURITY

Plaintext: TALHACRICKET

Diagram :-

[TA, LH, AC, RI, CK, ET]

Matrix :-

S	E	C	U	R
I	T	Y	A	B
D	F	G	H	K
L	M	N	O	P
Q	V	W	X	Z

:- Encryption :-

i) TA

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

T → Y

A → B

Ciphertext: US

DAY: _____

DATE: _____

(iii) Y U

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

Y $\rightarrow$ I A

U $\rightarrow$ ~~C~~ C

Decrpt :

(iv) SB

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

S $\rightarrow$ ~~C~~ R

B $\rightarrow$ ~~I~~ I

Decrpt :

(v) R G

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

R $\rightarrow$ ~~C~~ C G $\rightarrow$ K

Decrpt : CK

Decryption :-

Diagram = [YB, OD, YU, SB, RG, TE]

i) YB

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

Y → T

B → A

Decrypt : TA

ii) OD

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

O → L

D → H

Decrypt : LH

DAY: _____

DATE: _____

(vi) T F

	0	1	2	3	4
0	S	E	C	U	R
1	T	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

T $\longrightarrow$ E

F $\longrightarrow$ T

Decrypt : E T

Decrypt text : T A L H ^A C R I C K E T

DATE: _____

DATE: _____

ii) LH

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

L → O

H → D

Ciphertext : OD

iii) AC

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

A → Y

C → U

Ciphertext : YU

iv) RI

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

R → S

I → B

Ciphertext : SB

DAY: _____

DATE: _____

(v) CK

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

C → R
K → G

Cipher Text : RG

(vi) ET

	0	1	2	3	4
0	S	E	C	U	R
1	I	T	Y	A	B
2	D	F	G	H	K
3	L	M	N	O	P
4	Q	V	W	X	Z

E → T
T → F

Cipher Text : TF

Cipher Text : VBODYUSBRGTF

Testing Code

Playfair Matrix:

SECUR

ITYAB

DFGHK

LMNOP

QVWXZ

Plaintext: TALHACRICKET

Ciphertext: YBODYUSBRGTF

Decrypted: TALHACRICKET